

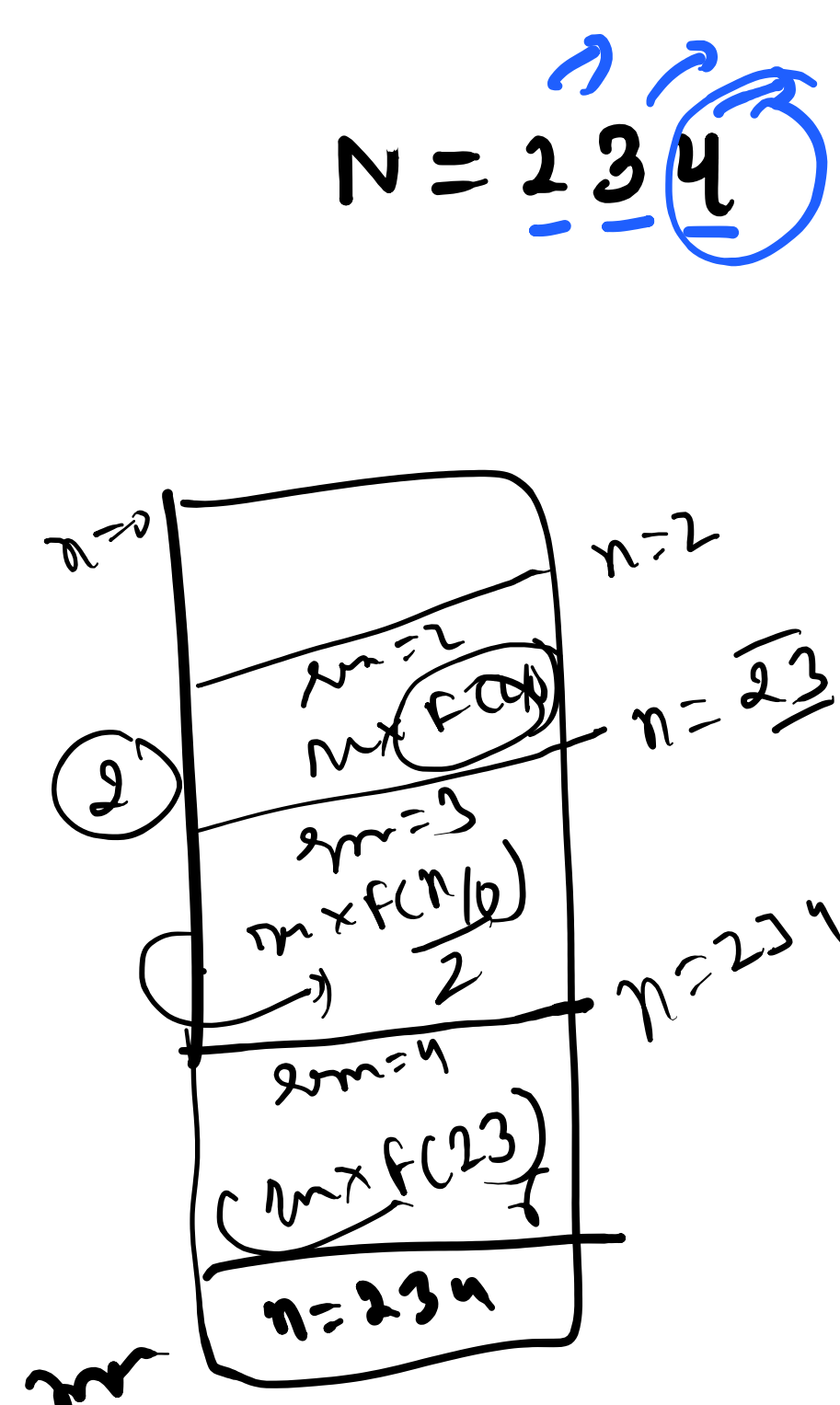
Q7. Product of digits

Example:

Example:
Input: N = 234

Input: N =
Output: 24

Q8. B



$$f \subset \pi^2 \Sigma$$

$$\text{if } cn = \text{vol}$$

$$\sim 1$$

$$2em = \pi \times 10$$

$$1 + 2em \times f(cn/10)$$

1×10

$\frac{12}{3}$

$n=3$
 $f(n=9, 123)$

$n=2$
 $f(n=3, 12)$

$n=1$
 $f(n=3, 1)$

$n=321$

$n=3$
 $n=12$

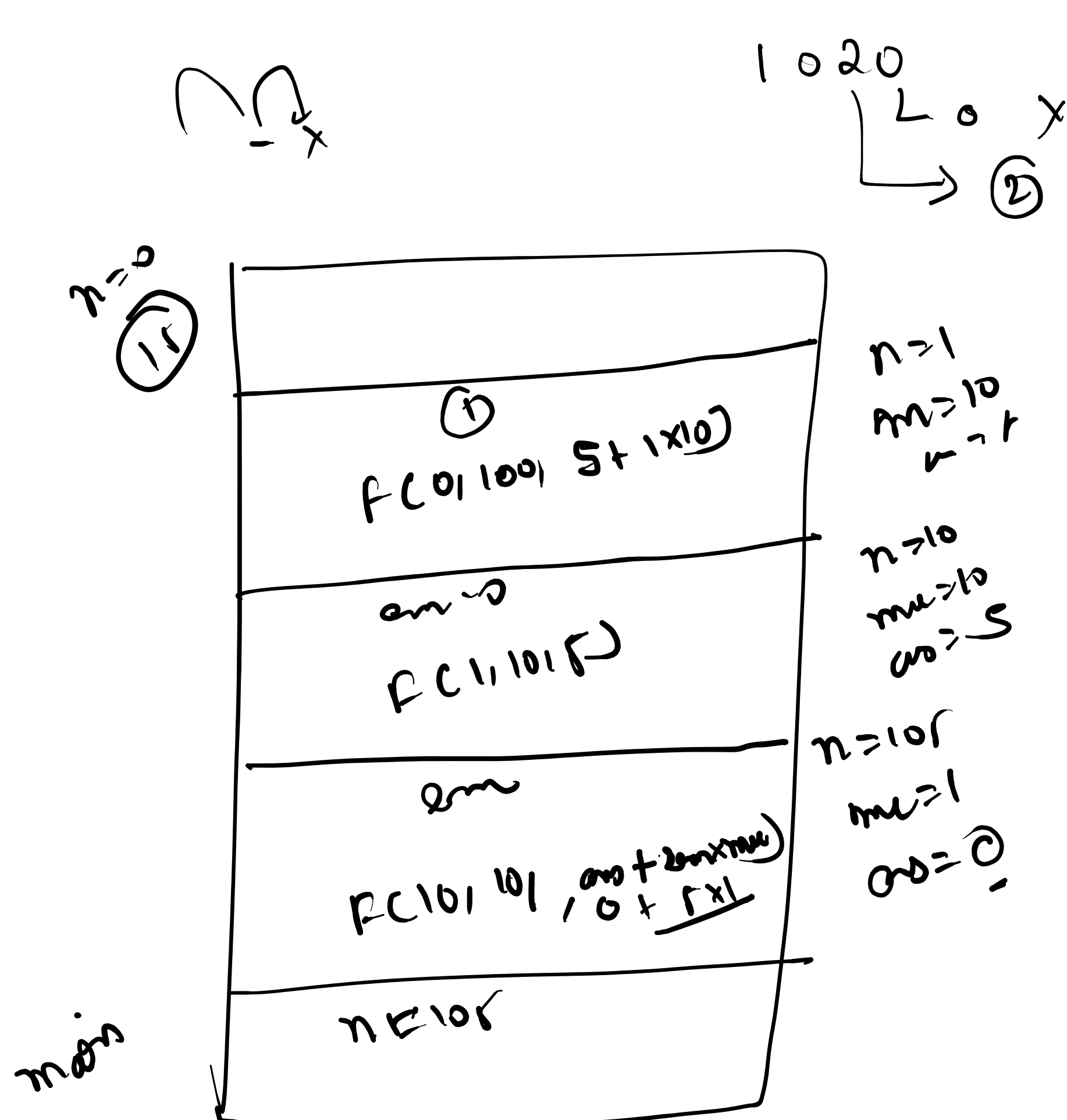
$n=32$
 $n=1$

man

$$f(n, \text{rev}) \begin{cases} \text{if } n == 0 \\ n = 0 \\ \text{rev} = n \times 10 \end{cases}$$

$$P(r, u)$$

given x and t rem



$$f(n) = n, mod, am)$$

$$2m = n \cdot 10$$

$$10 \cdot 10m = 10 \cdot 10$$

$$f(n/10)$$

$$3$$

$$e^2$$

$$f(n/10, mod, am) \quad \text{am + mod}$$

$\text{col} \rightarrow$
 $\text{at} \rightarrow \text{c}$
 $\text{t} \rightarrow \text{co}$
 $\rightarrow \text{cot}$

PCS_1 on Σ
 $\text{char}(ch) = \text{S.char}(ci) \wedge$
 $\text{if}(ch = 'a') \wedge$
 $\text{RCS.subst}(ci, \text{out}'b')$
 3
 else
 $\text{R.sub}(ci, \text{out}(ch))$

$$\begin{array}{r} 1234 \\ \hline 123 \\ \hline 1 \end{array}$$

$N=0$

0	1	1	2	3	5	8	13	-	-	-
0	1	2	3	4	5	6	9			

$$f(n) = \begin{matrix} f_1 = \text{fib}(n-1) \\ f_2 = \text{fib}(n-2) \\ \vdots \\ p_{k-1} \end{matrix}$$

$(n-1) \quad n$

$0 \quad 1 \quad 2 \quad 3 \quad 5 \quad 8 \quad 13 \quad 21 \quad 37$

$(n+1)$

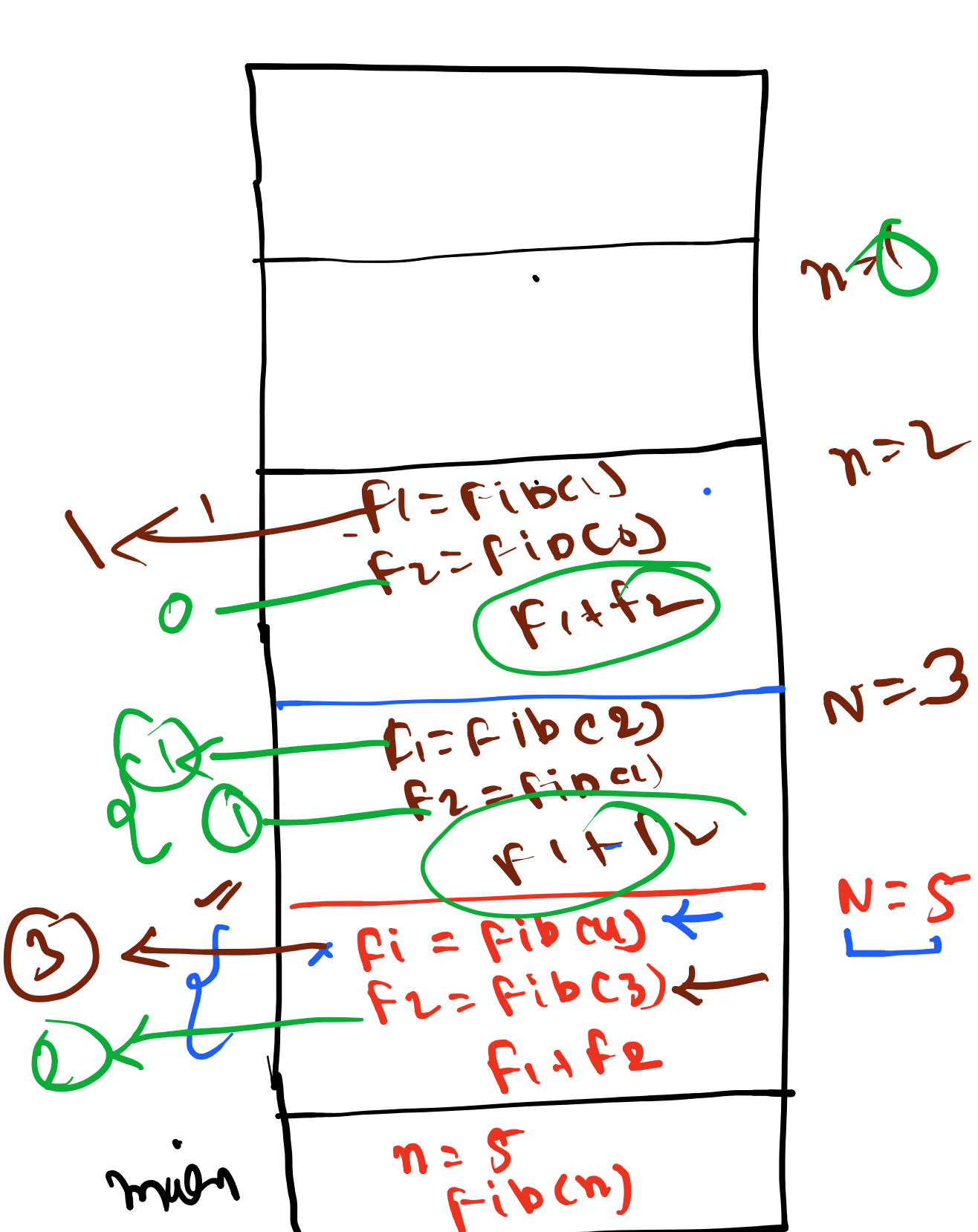
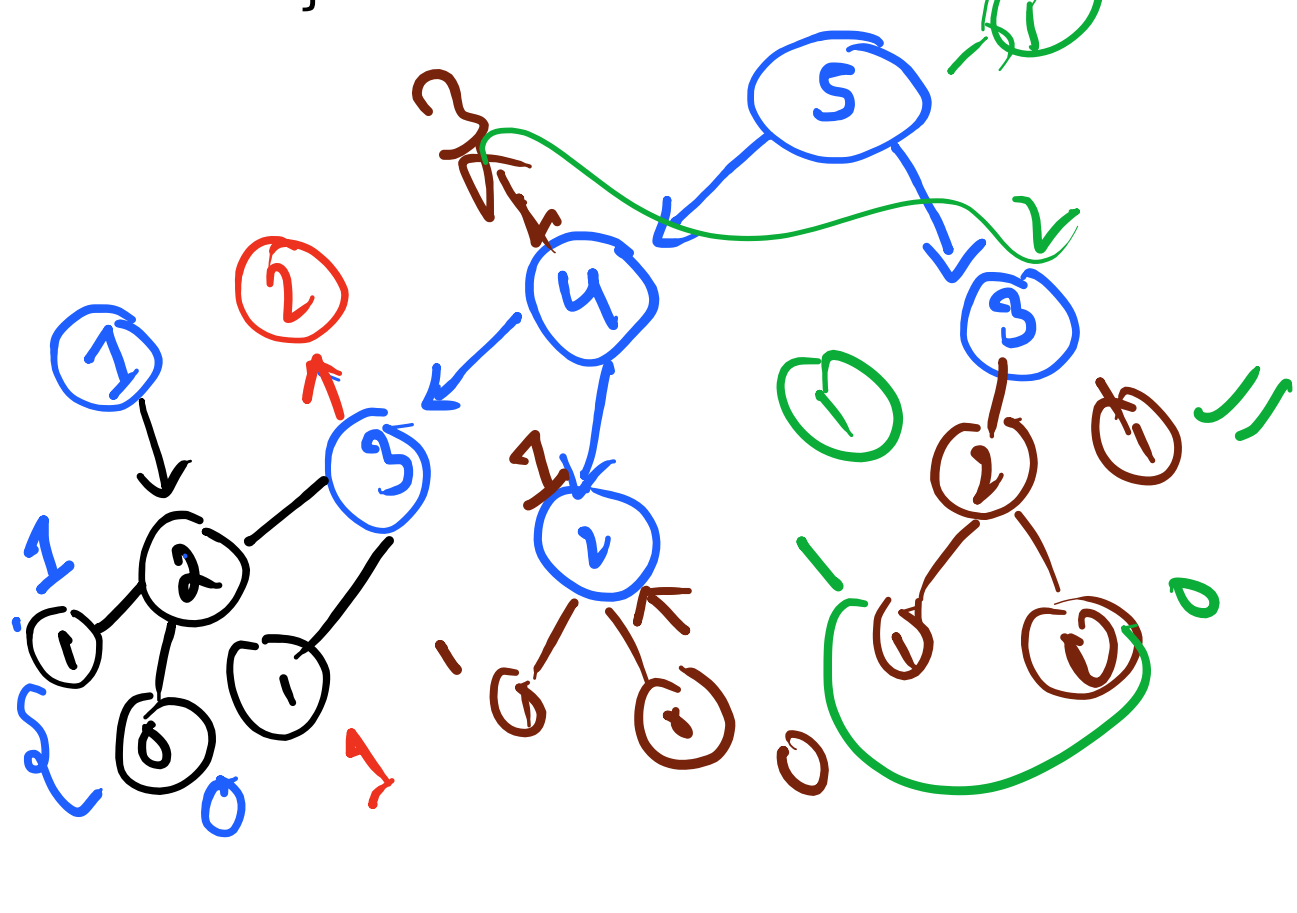
fib(n) 2 if (n == 0 || n == 1) {

$$f_1 = \text{fib}(n-1)$$

$$f_2 = \text{fib}(n-2)$$

$$f_1 + f_2$$
}

```
public static int fib(int n) {
    if (n == 0 || n == 1) {
        return n;
    }
    int f1 = fib(n - 1);
    int f2 = fib(n - 2);
    return f1 + f2;
}
```



```
public static int fib(int n) {  
    if (n == 0 || n == 1) {  
        return n;  
    }  
    int f1 = fib(n - 1);  
    int f2 = fib(n - 2);  
    return f1 + f2;  
}
```

